Jonathan Jacob Koshy

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Education

University of Waterloo Waterloo, ON
BASc. Electrical Engineering Expected 2028

Skills

- **HDL/Digital Design:** Verilog, SystemVerilog, RTL Design, FSMs, CDC, ASIC Design, FPGA
- **Verification/EDA Tools:** Cocotb, Icarus Verilog, GTKWave, GitHub Actions, Makefile
- **Programming:** C, C++, Python, RISC-V Assembly, MATLAB, Git
- **Protocols/Platforms:** SPI, I2C, UART, BLE, CAN, BACnet, PWM, STM32, ESP32, Raspberry Pi, Linux

Experience

Firmware Development Intern – Belimo - Montreal, QC May 2026 – Aug 2026

- Developed bare metal **Embedded C** firmware for memory constrained HVAC sensor and actuator controllers
- Implemented interrupt driven control logic using global state, binary flags, and deterministic firmware routines
- Validated **CAN, UART, and I2C** traffic to debug controller state machines, sensor reads, and actuator commands
- Verified PCB hardware and **BACnet** behavior across sensor readings, actuator responses, and integration tests

ASIC Digital Member - UWASIC – Waterloo, ON Jan 2026 – Present

- Designed **RTL** for an SPI controlled PWM peripheral using Mode 0 SPI transactions and memory mapped registers
- Implemented clock domain crossing, edge detection, bit counting, and address validation for SPI inputs
- Verified PWM frequency, duty cycle accuracy, and register behavior using Cocotb, Icarus Verilog, and GTKWave

Embedded Systems Intern – AeroCardia - Montreal, QC Sep 2025 – Dec 2025

- Implemented sensor drivers for IMU, PPG, and temperature sensors in C++ enabling real time monitoring
- Built a FreeRTOS pipeline streaming biosensor data under **50 ms latency** with less than 1% BLE packet loss
- Enabled firmware updates by implementing secure **BLE OTA** with image verification and rollback support
- Redesigned ESP32 analog signal chain using external ADC and the PCB layout reducing biosensor noise by **30%**

Embedded Flight Systems Member – UWARG – Waterloo, ON Dec 2024 – Sep 2025

- Implemented roll and yaw mixing firmware enabling coordinated control across **4+ motors** in fixed flight modes
- Developed **PID** attitude stabilization improving pilot control response and reduced oscillation during manual flight
- Built a motor and ESC validation test platform enabling safe integration testing before flight deployment

Firmware Member – Electrium Mobility – Waterloo, ON May 2025 – Aug 2025

- Built event driven **BLE** communication on **ESP32** enabling non blocking telemetry and stable control loop timing
- Integrated **VESC** telemetry over **UART** streaming RPM, current, and battery metrics for real time diagnostics

Information Technology Intern – University of Waterloo ECE Department - Waterloo, ON Sep 2024 – Dec 2024

- Diagnosed OS, driver, and hardware failures across **100+ Linux and Windows** lab machines used for research
- Built and configured workstation PCs including CPU, GPU, memory, storage and PCIe devices

Projects

Edge AI Dementia Companion (HackCanada Winner) *Python, YOLOv8, OpenCV, TypeScript, RPi 5*

- Built an edge AI assistant on **RPi 5** using **Hailo-8L** accelerated **YOLOv8** detection
- Added medication verification and object memory using **OpenCV** barcode scanning and tracking

ESP32 Wi-Fi Media Controller | *C++, ESP32, LVGL, Wi-Fi, REST APIs* | |

- Built an **ESP32** touchscreen interface using **LVGL** for metrics and media playback
- Streamed Spotify metadata and artwork while enabling remote control through Discord

RISC-V Interrupt Driven Embedded Controller | *RISC-V Assembly, SiFive FE310, Interrupts*

- Built a real time ISR capturing pseudo random values for an LED countdown timer
- Configured **PLIC**, **GPIO interrupts**, **mtvec**, **mie**, and **mstatus** registers